

Percentage Change

AQA · KS3 Year 7

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE



$$\frac{1}{2} = 0.5 = 50\%$$

shaded part = 1 of 2 equal parts = half

One half shown three equivalent ways.

COMMON MISTAKES

Dividing by the NEW value — always divide by the ORIGINAL.

Adding the percentage of the wrong amount.

20% increase new value is 120% of original, not 20% of original.

15% off then 15% on does NOT get back to the start.

METHOD

- 1 Find the percentage of the original value
- 2 Increase: add it on
- 3 Decrease: take it off
- 4 Percentage change = $(\text{change} \div \text{original}) \times 100$
- 5 State increase or decrease

WORKED EXAMPLE

A coat costs £80, reduced by 15%. Find the sale price.

Step 1 — 15% of £80 = £12

Step 2 — Decrease: £80 - £12 = £68

Sale price = £68

VISUAL AID

Bar model: Full bar = original amount (100%). Shaded portion = percentage kept. Unshaded portion = amount removed.

Formula: Percentage change = $(\text{change} \div \text{original}) \times 100$.

1. (2 marks)

Find: a) 10% of £50 b) 25% of £120 c) 5% of £200 d) 1% of £350

2. (2 marks)

Increase: a) £40 by 10% b) £60 by 25% c) £200 by 50%

3. (2 marks)

Decrease: a) £80 by 20% b) £150 by 10% c) £500 by 35%

4. (3 marks)

TV price rises from £450 to £495. a) How much increase? b) Percentage increase?

5. (3 marks)

Bicycle £320, 15% off. a) Discount? b) Sale price? c) +15% back on — is it £320? Explain.

6. (3 marks)

Factory: Jan 250 units, Feb 300 units. Calculate percentage increase.

7. (4 marks)

Mo: buys £60, sells £78. Abi: buys £45, sells £54. a) Mo's % profit. b) Abi's %. c) Who did better?

8. (5 marks)

House £200k. +12% Year 1, -8% Year 2. a) Value after Y1? b) After Y2? c) Overall % change?